

NING Siyuan

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EDUCATION

Master of Science in Physics

2023.08-2024.07

National University of Singapore

Relevant Coursework: Computational Condensed Matter Physics, Numerical Recipes with Applications, Bayesian Statistics and Machine Learning, Applied Machine Learning, Complex System Analysis and Modelling, Physics of Nanostructures, Thin Film Physics and Technology.

Bachelor of Science in Physics, minor in Artificial Intelligence

2019.09-2023.06

Shandong University

Relevant Coursework: Computer Modelling of Materials, Computational Physics, Group Theory, Solid State Physics, Film Physics and Techniques; (**minor**) Machine Learning, Neural Network and Deep Learning, Natural Language Processing.

RESEARCH EXPERIENCE

MSc Physics Project supervised by Prof. QUEK Su Ying, National University of Singapore

2023.08-2024.07

Worked on first-principles studies of quantum defect candidates in 2D WS₂

- Developed a workflow to investigate 48 substitutional defects in monolayer WS₂ by determining their ground-state structures, defect stability, spin configurations, and defect level positions.
- Identified three spin-triplet defects, and examined their optimized local geometries as well as electronic properties.
- Expedited setup and post-processing of calculations using bash scripts.
- Wrote a 60-page project thesis and present the results to a panel of examiners [\[PDF\]](#).

Research Assistant under Prof. Zhang Peng, Shandong University

2020.06-2022.11

Worked on computational analysis of hydrogen bond vibrations of ice III in the far-infrared band

- Calculated vibrational spectra of ice III and its hydrogen-ordered counterpart ice IX, validated by experimental data.
- Analyzed vibration normal modes of hydrogen bonds to explain the origin of far-IR vibrational bands.
- Assessed structural deformations under high pressure in hydrogen bond lengths and included angles.
- Authored first-authored paper published in *Crystal* [\[PDF\]](#).

RELEVANT PROJECTS

Machine learning in Materials Science

2024.07

VASPKIT Workshop, Tencent Quantum Laboratory

- Trained and fine-tuned advanced ML models, including Kernel Ridge Regression, Random Forest and XGBoost.
- Employed physics-informed featurization (Matminer) and Interpretable Machine learning techniques (SISSO, SHAP).

Sunspot Number Forecasting: Bayesian Neural Networks (BNNs) via MCMC [\[HTML\]](#)

2024.05

Final Project for PC5251 *Applied Machine Learning*, National University of Singapore

- Implemented stochastic gradient Langevin dynamics (SGLD) to find posterior distributions of model parameters.
- Obtained predictions with confidence intervals and conducted convergence analysis over different sample sizes.

Deep Learning the Phase Transition: A CNN Approach to the 2D Ising Model [\[HTML\]](#)

2023.10

Research Project in PC5212 *Numerical Recipes*, National University of Singapore

- Used the Monte Carlo method to obtain representative sets of spin configurations for a bunch of temperatures.
- Trained convolutional neural networks via PyTorch to classify configurations and predict the critical temperature.

RESEARCH SKILLS

Computational Modeling Codes/Packages:

- DFT codes: VASP, CASTEP, SIESTA
- Automation & Processing: VASPKIT, Pymatgen
- Model Builder and Visualizer: Material Studio, VESTA
- Finite Element Analysis: ANSYS Workbench

Machine Learning Techniques:

- Frameworks: Scikit-learn, PyTorch, XGBoost, SISSO
- Ensemble methods: Bagging, Boosting, Stacking
- Explainable tools: SHAP

Programming Language: Python (Numpy, Pandas), Shell (bash), MATLAB

Data Processing: Excel, Origin

VOLUNTEER & TEACHING EXPERIENCE

Volunteer Coordinator | Students' Union, School of Space Science and Physics, Shandong University **2020-2021**

Led and coordinated volunteer activities at nursing homes and schools for children with autism in Weihai.

Teaching Assistant | Teaching Office of General Mathematics, Shandong University **2020-2021**

Assisted in *Calculus* courses by preparing course materials, conducting tutoring, and grading.

GRANTS & AWARDS

Second-class Academic scholarship, Shandong University 2021-2023

Excellent Social Practice Research Report, Shandong University 2021

Volunteer Service Award, Shandong University 2020

PUBLICATION

Ning, S.-Y.; Cao, J.-W.; Liu, X.-Y.; Wu, H.-J.; Yuan, X.-Q.; Dong, X.-T.; Li, Y.-N.; Jiang, Y.; Zhang, P. *Computational Analysis of Hydrogen Bond Vibrations of Ice III in the Far-Infrared Band*. Crystals 2022, 12, 910. <https://doi.org/10.3390/cryst12070910>